Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) 3rd Year, 2nd Semester Final Examination-2024

Course Code: EEE 337 **Session:** 2021-2022

Course Title: Power System II

Total Marks: 100 **Credits: 3.0** **Total Time: 3 Hours**

[The figures in the right margin indicate full marks]

Part-A

*[Answer any **TWO** of the following questions]*

1. (a) Define critical disruptive voltage. Write down it's significance. 2

   (b) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. 5

   (c) Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing. 8

   (d) A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero. 10

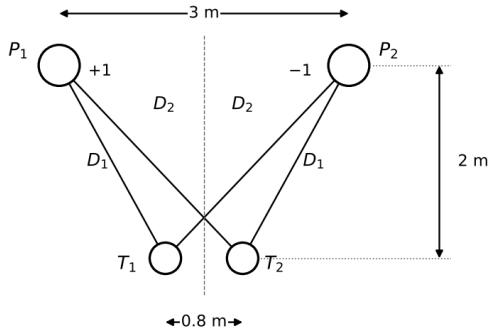


Fig. 1 of the paper (redrawn): figure for Q-1(d)

2. (a) What is monopolar link? 2

   (b) Write down the general construction of a 3-conductor cable with required figure. 5

   (c) A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current. 8

   (d) Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath. 10

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3. (a) How electrical breakdown happen in a pin-type insulator? 2

(b) What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)? 5

(c) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency. 8

(d) Derive the mathematical expression of String efficiency with proper diagram. What will happen if:

(i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$)

(ii) $K = 0.8$

(iii) the number of disc increases in the string. 10

Part-B

*[Answer any **TWO** of the following questions]*

4. (a) What is Stringing charts? 2

(b) The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is $4.5 \times 10^{14}\ \Omega - cm$, find the insulation thickness. 5

(c) Derive the expression of inductance of composite conductor lines. 8

(d) A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. 10

5. (a) How Earth affect on transmission line capacitance? 2

(b) A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line. 5

(c) A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively.

(i) Draw the equivalent circuit

(ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit

(iii) Calculate the line voltage and

(iv) Calculate string efficiency. 8

(d) Derive the expression of capacitance of a three phase line with unsymmetrical spacing. 10

6. (a) Find the most economical size of a single-core cable working on a 132 kV, 3- phase system, if a dielectric stress of 60 kV/cm can be allowed. 2

(b) Define HVDC. Discuss the Economics of Power transmission. 5

(c) Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram. 8

(d) The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($|V| = 1.0\ pu$), with the generator terminal voltage ($|V_t| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves. 10

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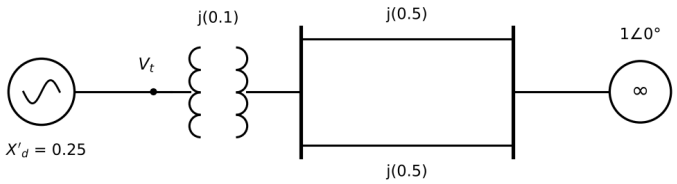


Fig. 2 of the paper (redrawn): figure for Q-6(d)

Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

[Term Test 1, Total Marks: 10, Time: 20 min]

7. (a) A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. 5

   (b) Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing.

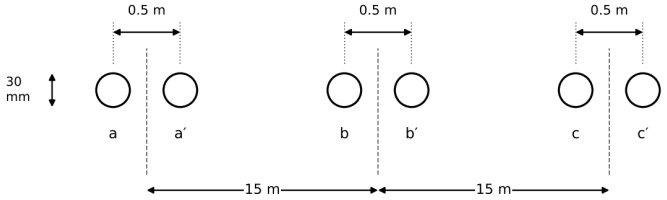


Figure 1 of Term Test 1 (redrawn)

Term Test 2 (its questions 1 and 2 are 8(a) and 8(b))

[Term Test 2, Total Marks: 10, Time: 20 min]

8. (a) A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. 5

   (b) A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 cm^2 and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current. 5

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Term Test 3 (its questions 1 and 2 are 9(a) and 9(b))

[Term Test 3, Total Marks: 10, Time: 20 min]

9. (a) In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. 5

   (b) A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. 5

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